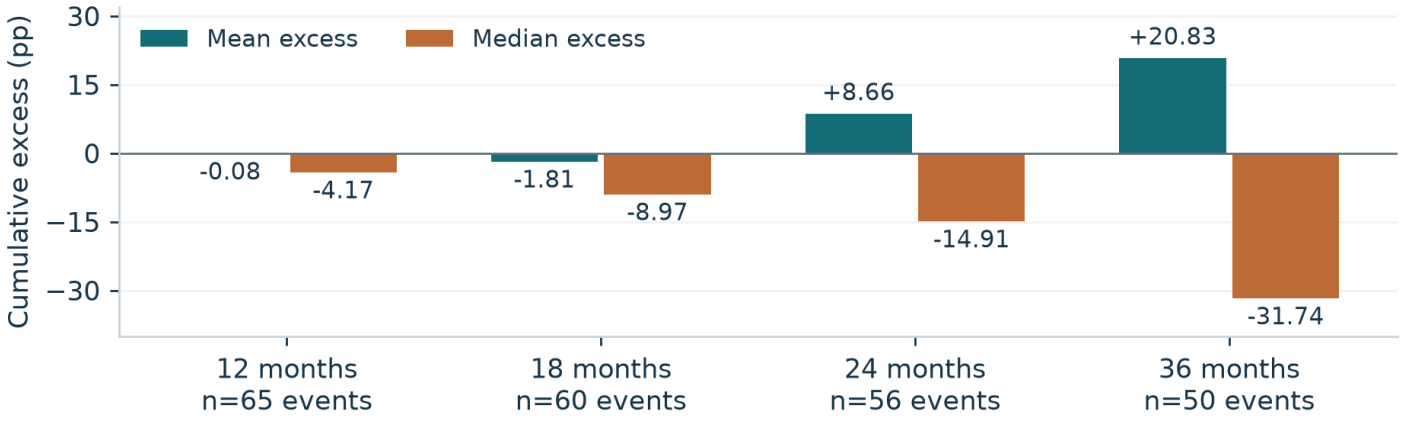
ML and NLP for long-term active positions

Discussion with Professor Michael Sternberg | Private Markets Modeling | Prepared October 4, 2026

Research aim: use machine learning (ML) and natural language processing (NLP) to uncover long-term alpha signals in companies backed by high-quality event-driven or activist hedge funds, then take active risk in several selected, potentially concentrated positions.

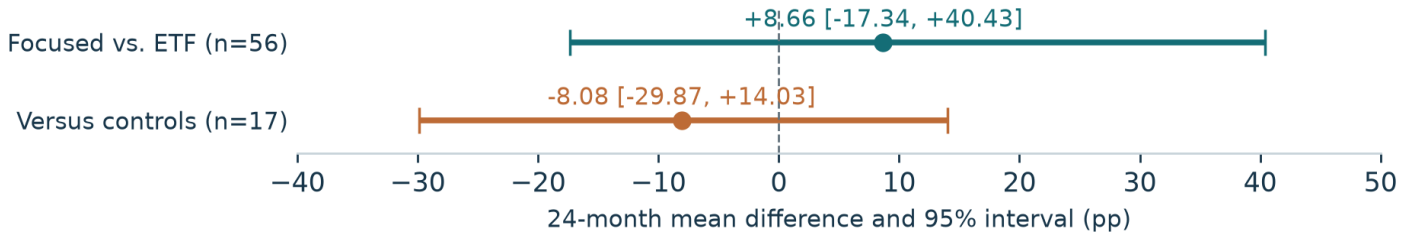
Current finding: simple following has not established a reliable investable edge. The historical baseline motivates a selection test, but does not show the winners were identifiable in advance.

1. Longer-horizon averages improve; the typical event still lags



Focused activist-candidate events; 2018-2025 exploratory research window. Different horizons have different, overlapping samples. These bars are gross cumulative stock-minus-Russell-size-ETF-proxy returns; pp = percentage points. [E1]

2. Two-year estimates remain uncertain



Dots show means; lines show saved issuer-bootstrap 95% intervals. Both cross zero. The 17 industry/year/size-matched cases are a subset with a different comparison group; do not pool the two estimates. [E1]

Two-year evidence	What it means
Mean +8.66 pp; median -14.91 pp 23 of 56 events beat the ETF (41.1%)	A positive average coexists with underperformance in most measured events. It is not a repeatable selection rule.
All events: 2,107 scheduled > 712 assigned > 694 calculable Focused: 67 scheduled > 58 assigned > 56 calculable	Coverage is selective. Missing observations may change conclusions; focused events sit inside the broader sample.

Conventions and limits that matter for interpretation

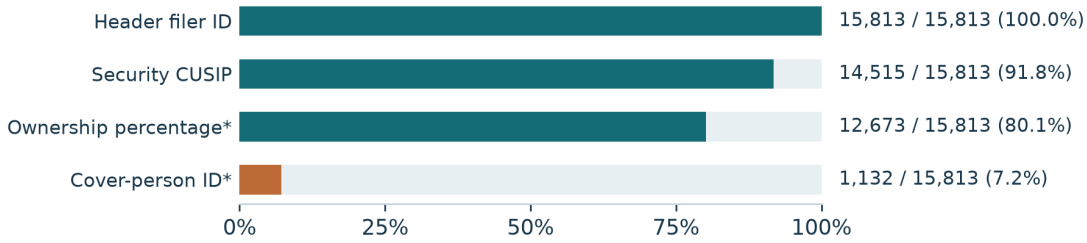
Modeled entry: opening price on the second exchange session after SEC acceptance; 252/378/504/756 sessions approximate 12/18/24/36 months. Events are equally weighted and gross of strategy costs; they are not funded-account or risk-adjusted alpha. Observed opening prices do not guarantee execution.

Funded simulations use 5% issuer targets, at most 20 issuers, liquidity limits, modeled costs, cash interest and no concurrent duplicate issuer positions. Primary $100k 18/24/36-month accounts remain incomplete at Aratana's replacement-share/contingent-right settlement. No sale proceeds are invented. Horizons were revised after earlier results were examined; this is exploratory development. [E1]

Research phases, current progress and data sources

Progression: simple following > structured ML selection > incremental NLP intent.

3. Filing data are prepared; economic identity remains a gap



2013-2025 selected initial-filing candidates. *At least one extracted value per filing; this does not imply every person or ownership row was recovered. Bars overlap and are not a sequential funnel. A CUSIP is not a verified historical stock link. [E2]

The three phases - what each adds

Phase / investment purpose	Information being tested	Current status
1. Simple following Establish a post-disclosure investment baseline.	Initial 13D events, manager groups and subsequent stock/benchmark returns.	Historical baseline evaluated; no reliable edge. Some funded accounts remain incomplete.
2. Structured ML Select stronger company/manager opportunities.	Manager history, ownership, size, liquidity, volatility and prior returns; fundamentals planned.	15,813 candidates audited; HFR scorecards built. Dated links/features unfinished; no fitted model.
3. Incremental NLP Test the added value of disclosed intentions.	Item 4 plans: board representation, capital return or sale; add to Phase 2 on matched samples.	Planned; no validated intent/alpha result. Identity/ownership pilot is preparation work.

Where the information comes from

Source	What it supplies	Use in this project
SEC EDGAR	13D/13D-A documents, acceptance times, filer/issuer identities, holdings and Item 4 text.	Cached filing archive; eligibility, factual extraction and future intent NLP.
CRSP via WRDS	Daily stock/ETF prices, returns, volume, security identifiers and delisting fields.	Market data for the reported historical return tests and simulations.
FTSE Russell via WRDS	Historical Russell constituent snapshots; benchmark returns use CRSP ETF data.	Assign stock-appropriate Russell size-segment ETF proxies.
HFR	35 supplied quarterly snapshots (2018 Q1-2026 Q3): fund/manager metadata, strategy, reported returns and assets.	Manager candidates and descriptive peer scorecards; not yet a tested selection signal.
Compustat via WRDS (planned)	Company fundamentals for valuation, leverage, profitability and balance-sheet features.	Access tested; a historically available ML feature panel is not complete.

WRDS = Wharton Research Data Services, the access platform; CRSP, FTSE Russell and Compustat are separate data sources. **Yahoo Finance** supplied an early ten-company price-only preview, not these reported return results; that script used Yahoo directly rather than yfinance. [E4]

HFR fund performance differs from returns available to an outside disclosure follower. Historical manager identity/availability and incomplete dead-fund coverage remain limits; today's rankings cannot become historical signals. [E3]

Next: validate dated inputs; freeze manager-quality and stock-ranking rules; test chronologically under common trading/cost conventions. Concentrated sizing remains proposed; baseline targets are 5%. Previously inspected history is development data.

Evidence: E1: long-horizon summary/controls, Sept 24. E2: field coverage/progress, Sept 25. E3: HFR protocol/scorecards, Sept 30; proposal V4. E4: market-source inventory and early preview scripts. Links in companion notes. AI-assisted synthesis; no new backtest or model fit.